

Case Study

Customer: “Meridian Home Services” (fictional, representative of our typical engagement)

Mid-market home-services company: HVAC, plumbing, and electrical. ~180 staff across 11 branch locations in 3 regions.

Channels: ~6,000 inbound calls/month + ~2,500 emails/month. A 9-person contact-center team books jobs, answers FAQs, and chases status updates.

Pain: long hold times, inconsistent answers across branches, agents spend ~40% of time on repetitive questions (hours, pricing bands, service-area checks, appointment status).

Goal: an AI assistant that resolves common requests end-to-end (answer policy/FAQ questions, check service-area eligibility, book or reschedule a visit) and cleanly hands off everything else to a human with full context.

Build a working prototype of the Meridian assistant. It should answer customer questions grounded in the provided knowledge pack, and complete at least one action against the mock Booking API. Use whatever LLM provider and frameworks you prefer (LangChain / LlamaIndex / LangGraph / none — your call; justify it).

Required capabilities:

1. Retrieval-grounded answering over the provided documents, with citations/traceability back to source.
2. An agentic flow that can check service-area eligibility for a ZIP and then create or reschedule a booking via the mock API — including input validation and a confirmation step before the action commits.
3. A safe fallback / human-handoff path when confidence is low, the question is out of scope, or required info is missing.
4. A small evaluation harness: a test set (you can extend ours) and metrics that show retrieval quality and answer/action correctness.

Deliverables

- Running code in a repo, with a README: how to run it, key design decisions, and what you deliberately left out.
- The eval harness + a short results summary.
- A one-page “path to production” note: what you would harden, monitor, and change to serve all 11 branches with branch-specific policies and ~8,500 interactions/month.
- Assumptions, debugging methodology, guardrails